MILITARY SPECIFICATION

CUTTER, POWDER ACTUATED, FACE CURTAIN CABLE, MARK 4 MOD 0

This specification is approved for use within the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the manufacture and acceptance of the Mark 4 Mod 0 powder actuated face curtain cable cutter (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

Military

MIL-P-116	Preservation, Methods of
MIL-W-5424	Wire Rope, Steel (Corrosion-Resisting) Flexible, Preformed
MIL-D-21625	Design and Evaluation of Cartridges for Cartridge Actuated Devices
MIL-P-46610	Primers, Percussion, Styphnate and Chlorate Types, for Small Arms Ammunition

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-453	Inspection, Radiographic

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Systems Engineering and Standardization Department (SESD) Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1377

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2.1.2 **Drawings and publications.** The following drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

Naval Air Systems Command (Code Ident 30003)

DL419690	Cutter, Powder Actuated, Face Curtain Cable, Mark 4 Mod 0 (Formerly Code Ident 10001)
DL537893	Container, Inner, for Cutter, Powder Actuated, Face Curtain Cable Mark 4 Mod 0 (Formerly Code Ident 10001)
DL537894	Container, Outer, for Cutter, Powder Actuated, Face Curtain Cable, Mark 4 Mod 0 (Formerly Code Ident 10001)

U. S. Army Munitions Command (Code Ident 19200)

7645339	Primer, Percussion (50M-5061W)

PUBLICATIONS

Naval Sea Systems Command (Code Ident 53711)

WR-43	Preparation of Quality Assurance Provisions
OP-5	Ammunition Ashore, Handling, Stowing and Shipping

Code of Federal Regulations

49 CFR 100-199	Transportation

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer).

2.2 **Order of Precedence.** In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 **First article.** When specified in the contract or order, a sample shall be submitted for first article inspection (see 4.4 and 6.3).

3.2 **Conformance to documents.** The Mark 4 Mod 0 powder actuated face curtain cable cutter covered by this specification shall be manufactured in accordance with DL419690 and all documents listed thereon.

3.3 **Materials.** All materials used in the manufacture of the cutters shall conform strictly to the specifications referred to on the respective drawings unless specific approval in writing covering a departure therefrom has been obtained from the cognizant Navy design activity prior to manufacture. When alternate materials or methods of manufacture are specified on the drawings, the bidder's selections shall be clearly stated in the proposal.

3.4 **Primary components.** For the purpose of this specification, the primer shall be considered a primary component (see 4.5.1).

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3.5 Primer. The primers shall be manufactured within 2 years before assembly of the primers into the cable cutters or within 1 year after recertification. The primers shall be the 50M type, manufactured in accordance with Drawing 7645339, loaded with primer mixture 5061W, and shall be furnished from lots which were accepted in accordance with MIL-P-46610, Class 1 (see 4.6.1).

3.5.1 Primer X-ray. All primers shall be free of the following defects: missing or abnormally thin charge; charge broken, cracked or porous; anvil cracked, off center, or abnormally close to the primer cup dome; and all other observable defects in assembly (see 4.6.2.1).

3.6 Cutter X-ray. Cable cutters shall be examined for missing parts and other imperfections in assembly (see 4.6.2.2).

3.7 Functional requirements.

3.7.1 Cable cutting. A steel cable of the type specified (see 4.6.4) shall be completely severed by each cable cutter firing.

3.7.2 Firing mechanism actuation. An applied pressure of 700 ± 25 pounds per square inch shall actuated the cable cutter firing mechanism.

3.7.3 Hangfire. The elapsed time between first movement of the firing pin to severance of the cable cutter shall not exceed 0.100 second. If the time exceeds 0.100 second it shall be considered a "hangfire".

3.7.4 Misfire. The primer shall fire as the result of firing mechanism actuation. If the primer does not fire it shall be considered a "misfire".

3.8 Workmanship. Cutters shall be free of the following defects: missing shear pin retaining screw, missing shear pin (if retaining screw is missing), threaded parts loose or improperly assembled, defective threads on the fitting end, obstruction of hole in fitting end, and any other defect which might prevent proper mating with connecting parts or proper functioning.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)
b. Quality conformance inspection (see 4.5).

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4.3 **Inspection conditions.** Unless otherwise specified (see 6.2), all inspections shall be performed as specified in the applicable test method of 4.6.

4.4 **First article inspection.** First article inspection shall be performed after award of contract and prior to production by a facility designated in the contract or order (see 6.2). First article inspection shall be performed on sample units which have been produced using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials, including packaging and packing shall be the same as used for regular production and shall be from the intended sources of supply.

4.4.1 **Sample size.** One hundred and eighteen cutters shall be submitted for first article inspection.

4.4.2 **Inspection routine.** One hundred and eight cutters shall be expended in the tests listed below and ten shall be retained for investigative purposes. A first article sample acceptable for environmental and functional testing shall be defined as a sample that has met the requirements of 3.2 through 3.6 of this specification. Sample acceptance shall consist of the following tests as prescribed in MIL-D-21625 (see 6.2), except that 160°F shall be used in place of 200°F for the high temperature and vibration tests: 6-foot drop (6); shock (12); temperature and humidity cycling (18); vibration (12); salt fog (12); high temperature exposure (12); normal temperature (12); low temperature (12); high temperature (12). The numbers in parenthesis denote the number of cutters allocated for each test. The cutters used for the low temperature and high temperature tests shall be conditioned at $-65° \pm 5°F$ and $160° \pm 5°F$, respectively, for not less than 6 hours nor more than 24 hours and fired immediately after removal from conditioning. All the other cutters shall be conditioned for not less than 6 hours at $70° \pm 5°F$ and fired immediately after removal from conditioning. All firings shall be conducted as specified in 4.6.4 and each of the cutters shall meet the requirements specified in 3.7.1 through 3.7.4.

4.4.3 **Failure.** Failure of the sample to pass the above tests shall be cause for rejection of the first article sample.

4.5 **Quality conformance inspection.** Quality conformance inspection shall consist of verification of the characteristics classified on the drawings of DL419690 and the tests and examinations of table I.

TABLE I. Quality conformance tests and examinations.

Item	Nature of Test	Reqmt. Para.	Test Para.	Classification (WR-43)
Primary Component	Single Lot Verification	3.4	4.5.1	(M101)
Primer	Age, loading and acceptance	3.5	–	(C1)
Primer	X-ray	3.5.1	-	(C2)
Cable cutter	Visual inspection	3.8	4.6.3	(M102)
Cable cutter	X-ray	3.6	4.6.2.2	(C3)
Cable cutter	Function requirements	3.7	4.6.4	(C4)
Cable cutter	Packaging	Section 5	4.5.4	(M103)

4.5.1 **Lot size.** Unless otherwise specified (see 6.2). the production lot size shall be not less than 500 and not greater than 1200 maximum. Cutters required for all test purposes are randomly selected from the production lot and shall not be applied as a part of the quantity specified for delivery by the contract. Only primary components (see 3.4) from a single lot shall be used in a production lot of cutters. One primary component production lot may be used in more than one cutter production lot.

4.5.2 **Sampling.** Sampling, with the exception of the ballistic sample, shall be in accordance with MIL-STD-105. The ballistic sample shall be randomly selected from each production lot in accordance with MIL-STD-105, general inspection level III, and an AQL of 0.10. Ten cutters from each production lot shall be furnished for investigative purposes.

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4.5.3 Classification of characteristics. The characteristics verified by the tests and examinations herein are classified as critical, major, or minor in accordance with WR-43. Tests and examinations that verify critical characteristics are identified by the symbol (C) and major characteristics by the symbol (M). The number following the classification symbol indicates the serial number of the test or examination. Tests and examinations which are not annotated with a classification code are classified as minor.

4.5.4 Preservation, packing and marking. The preservation and packing of the cutter and the container markings shall be in accordance with Section 5.

4.6 Tests.

4.6.1 Primer verification. The contractor shall provide verification of primer conformity with the requirements of 3.5.

4.6.2 Radiographic testing.

4.6.2.1 Primers. All primers shall be radiographically inspected in accordance with MIL-STD-453. Those having any of the defects specified in 3.5.1 shall be rejected. Pressure-sensitive tape may be used as an aid in positioning the primers in suitable grooved racks. For the most revealing exposure, the primers shall be laid on their side with the anvil cutouts vertical.

4.6.2.2 Cable cutters. All cable cutters shall be radiographically inspected in accordance with MIL-STD-453. Cutters having any observable imperfections in assembly shall be rejected. The cutters shall be positioned on their sides, with fitting end in a horizontal position, for the most revealing exposure. All cutters shall be identified with serial numbers prior to examination. The cutter serial numbers shall be in consecutive order beginning with the number 001 in each production lot. The cutters shall be arranged on trays or boards in consecutive numerical order, and each radiograph shall carry a permanent identification of the cutters displayed thereon. The radiograph identification shall include the complete lot number, as stamped on the cutters, and the span of serial numbers displayed. Discontinuities in serial numbers shall be identified on the X-ray and those defectives removed from the production lot. The serial numbers shall be pen or ink stamped on the cutters with approximately 1/8-inch-high characters. The serial numbers shall be located apart from the other cutter markings.

4.6.3 Visual inspection. All cable cutters shall be visually inspected. Those having any of the defects specified in 3.5.1 shall be rejected.

4.6.4 Ballistic test. The cable cutters shall be temperature conditioned for not less than 6 hours at 70° ± 5°F before firing. The cable to be cut by the cable cutter firings shall be flexible steel wire rope in accordance with MIL-W-5424, 1/16-inch diameter, 7 × 7 construction. Before firing, a new piece of the cable shall be positioned for cutting in the slotted end of each cable cutter. The firing mechanism shall be actuated by 700 ± 25 pounds per square inch of propellant gas pressure or air pressure applied through the fitting end of the cable cutters. Suitable instrumentation shall be used on each firing to record the elapsed time between application of the actuating pressure to the firing mechanism and cutting of the cable.

5. PACKAGING

5.1 Preservation and packing. For definitions of packaging levels see 6.5.

5.1.1 Level A. Level A packaging shall be used for packaging of all production lots for service use. Two cutters shall be packaged in an inner container in accordance with the drawings listed on DL537893.

5.1.2 Level C. When directed by the procuring activity (see 6.2), cutters shall be packaged in accordance with Method III of MIL-P-116 and with 49 CFR 171-179.

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5.2 Packing. For definitions of packing levels see 6.5.

5.3 Marking.

5.3.1 Special marking. Marking of outer containers shall be in accordance with 49 CFR 171-178.

5.2.2 Normal marking. Unless otherwise specified in the contract or order, the marking information on unit packages and shipping containers shall be as specified on the drawings listed on DL537893 and DL537894, respectively. Marking information shall be applied to the containers in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The Mark 4 Mod 0 powder actuated cutter is intended to sever the face curtain cables after seat ejection from T-2 or A-5 aircraft.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number and date of this specification
b. Whether first article inspection is required and if so, specify the test activity
c. That revision "D" of MIL-D-21625 be used
d. Inspection conditions, if other than as specified (see 4.3)
e. Production lot size, if other than as specified and test activity (see 4.5.1)
f. Special marking, if any (see 5.3.1)
g. Level of preservation and packing (see 5.1 and 5.2)
h. That the safety precaution requirements of the "Contractors' Safety Manual for Ammunition, Explosives, and Related Dangerous Material," DOD 4145.26M, are applicable. NOTE: When this specification is used as part of the description of work to be accomplished by a Government Activity, the safety precaution requirements of "Ammunition and Explosives Ashore," OP 5, are applicable.

6.3 First article. When a first article inspection is required, the item should be a first article sample. The first article should consist of 118 cartridges. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirements for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Definitions. For the purposes of this specification, the following definitions apply.

6.4.1 Level A packaging. The degree of preservation-packaging which will afford adequate protection against corrosion, deterioration, and physical damage during handling, shipment, indeterminate storage, and worldwide redistribution.

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6.4.2 Level C packaging. The degree of preservation and packaging which will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity, for immediate use. This level may conform to supplier's commercial practice when such meets the requirements of this level.

6.4.3 Level A packing. The degree of packing which will afford adequate protection during shipment, handling, indeterminate storage, and world-wide redistribution.

6.4.4 Level C packing. The degree of packing which will afford protection against damage during direct domestic shipment from the supply source to the first receiving activity for immediate use. This level in general will conform to the applicable carrier rules and regulations, and may be the supplier's commercial practice when such meets the requirements of this level.

Preparing Activity
NAVY-AS
(Project 1377-N781)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-23288C(AS)	CUTTER, POWDER, ACTUATED, FACE CURTAIN CABLE, MARK 4 MOD 0

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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